

ReDataX - Validation, Robustness and Reproducibility

Validation, Robustness and Reproducibility

This dashboard documents the chronological validation protocol, model-selection governance and robustness of the final results.

Train: 6-26 January 2025 **Development:** 27 January - 2 February 2025 **Validation:** 3-9 February 2025 **Holdout:** 10-16 February 2025

The model configuration and horizon were selected before the final holdout was evaluated.

Confidence intervals are based on paired day-level bootstrap. They describe variation across seven daily clusters, not independent even t-level uncertainty.

Results represent scenario-adjusted potential protected value, not realized bank PnL.

Temporal Validation Protocol

Split	Start Date	End Date	Days	Purpose	Allowed Decisions	Forbidden Decisions
train	2025-1-6	2025-1-26	21	Fit model parameters	Model fitting	Policy selection and final reporting
development	2025-1-27	2025-2-2	7	Select model and policy within each horizon	Model preset, policy threshold and budget selection	Final horizon confirmation and final reporting
validation	2025-2-3	2025-2-9	7	Select one horizon and confirm robustness	Horizon selection among development winners	Retuning on final holdout
final	2025-2-10	2025-2-16	7	One-time evaluation of the frozen configuration	Reporting only	Any parameter or threshold tuning

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Development and Validation Selection Path

Market	Stage	Horizon, sec	Leaderboard Rows Stored	Accepted Among Stored	Outcome	Selected Model	Budget	Margin, bps	Min P(BE)	Mean Daily Value / \$1M	Robust Score / \$1M	Positive Days
BTCUSDT	development	120	20	20	Selected	compact	10%	0.1	40%	5.41	3.57	100%
BTCUSDT	development	300	20	20	Selected	medium	10%	0.1	40%	10.83	8.23	100%
BTCUSDT	development	600	20	20	Selected	medium	10%	0.1	40%	15.77	13.29	100%
BTCUSDT	validation	120	1	1	Rejected	Rejected	Rejected (0%)	Rejected, (0%)	Rejected, (0%)	Rejected (0)	Rejected (0)	Rejected (0%)
BTCUSDT	validation	300	1	1	Rejected	Rejected	Rejected (0%)	Rejected, (0%)	Rejected, (0%)	Rejected (0)	Rejected (0)	Rejected (0%)
BTCUSDT	validation	600	1	1	Selected	medium	10%	0.1	40%	16.8	9.15	100%
ETHUSDT	development	120	20	20	Selected	compact	10%	0.1	Rejected, (0%)	14.67	9.55	100%
ETHUSDT	development	300	20	20	Selected	compact	10%	0.1	40%	21.77	11.35	100%
ETHUSDT	development	600	20	20	Selected	medium	10%	0.1	Rejected, (0%)	28.7	18.69	100%
ETHUSDT	validation	120	1	1	Rejected	Rejected	Rejected (0%)	Rejected, (0%)	Rejected, (0%)	Rejected (0)	Rejected (0)	Rejected (0%)
ETHUSDT	validation	300	1	1	Rejected	Rejected	Rejected (0%)	Rejected, (0%)	Rejected, (0%)	Rejected (0)	Rejected (0)	Rejected (0%)
ETHUSDT	validation	600	1	1	Selected	medium	10%	0.1	Rejected, (0%)	30.33	20.6	100%

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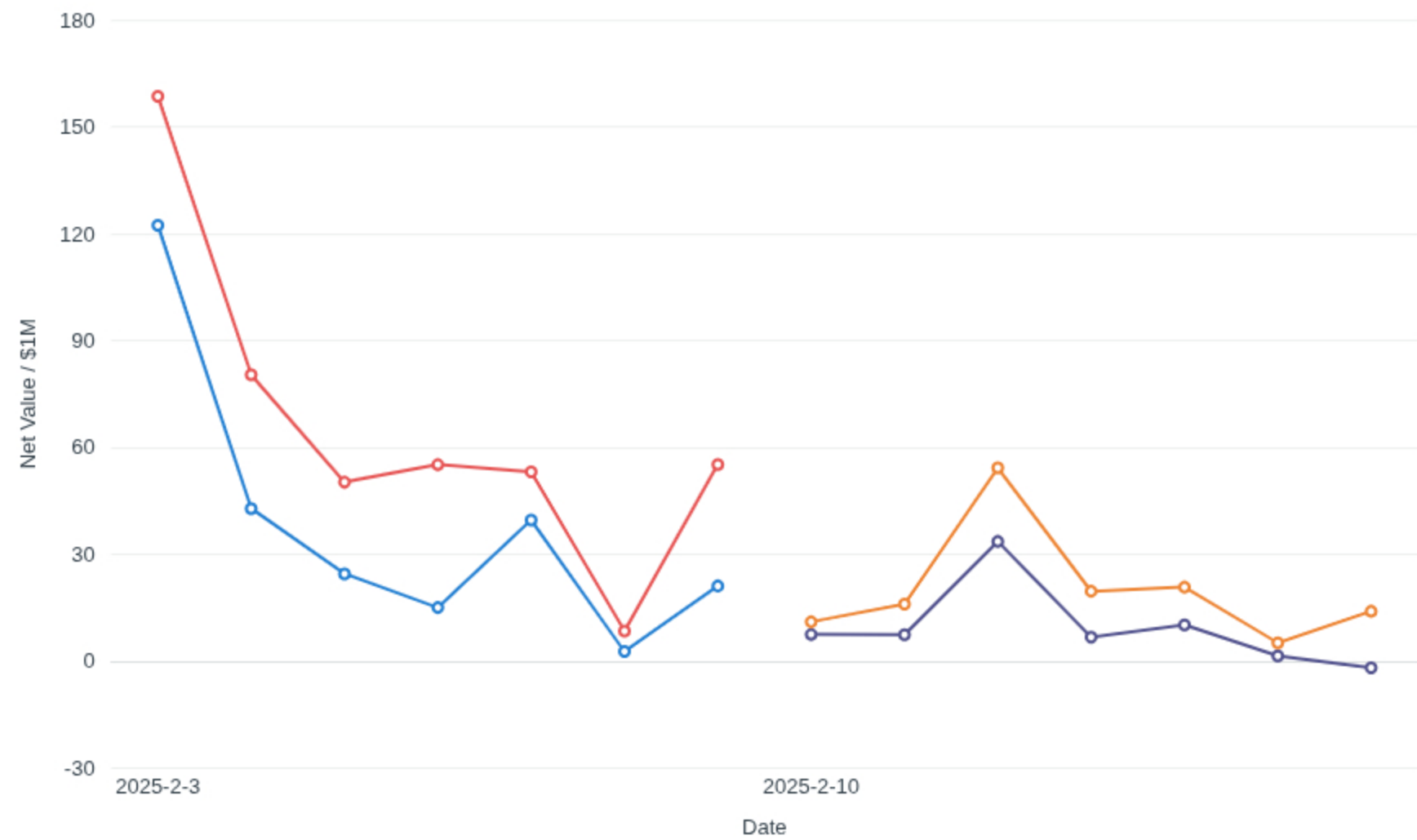
Validation To Final Consistency

Market	Horizon, sec	Validation Mean / \$1M	Validation Std / \$1M	Validation Positive Days	Final Mean / \$1M	Final Std / \$1M	Final Positive Days	Final - Validation / \$1M	Direction Consistent	Final Majority Positive
BTCUSDT	600	16.8	15.29	100%	9.25	10.62	85.7%	-7.54	Yes	Yes
ETHUSDT	600	30.33	19.46	100%	20.08	14.77	100%	-10.24	Yes	Yes

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Daily Hurdle Value: Validation And Final

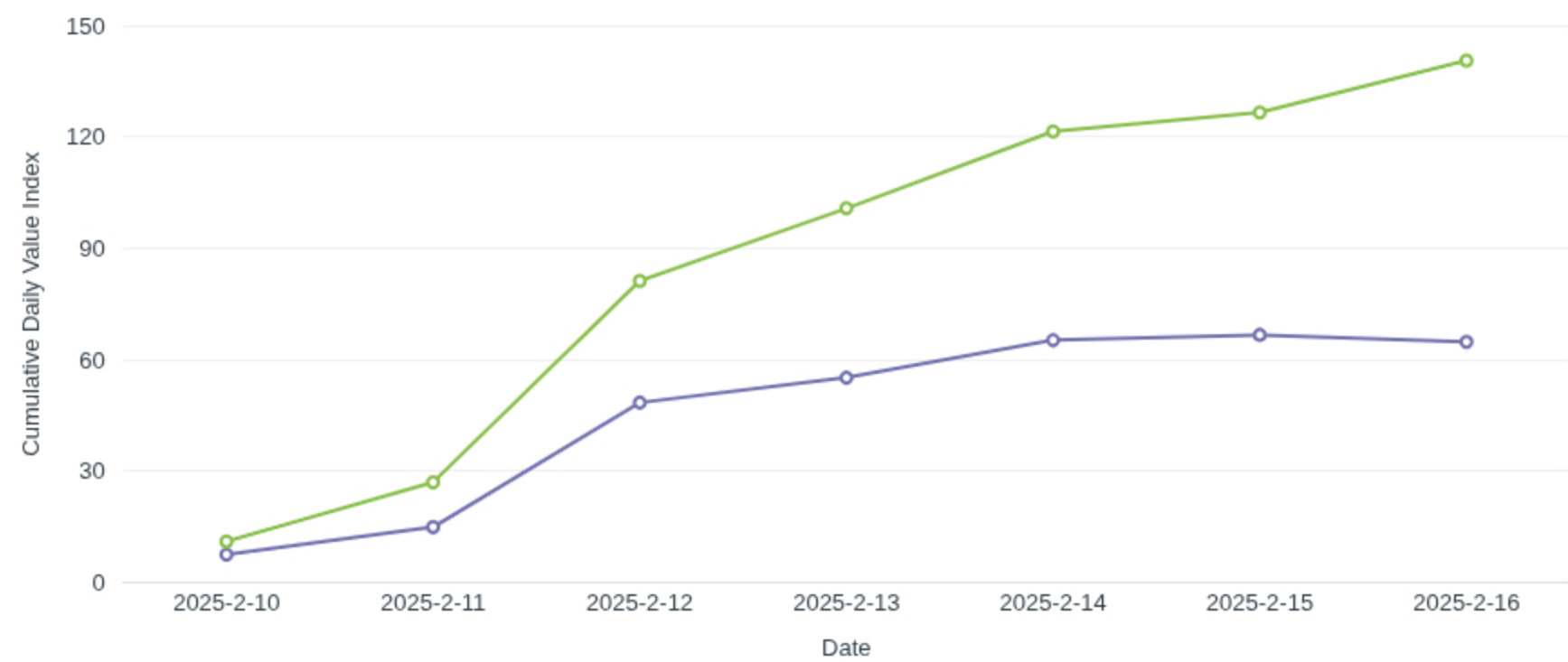
● BTCUSDT | validation ● ETHUSDT | validation ● BTCUSDT | final ● ETHUSDT | final



The cumulative series is the running sum of equal-weighted daily normalized values. It's a robustness index, not cumulative realized bank profit.

Cumulative Final Daily Value Index

● BTCUSDT ● ETHUSDT



Daily Robustness by Policy And Split on Horizon 600 sec									
Split	Market	Policy	Days	Mean / \$1M	Median / \$1M	Std / \$1M	Worst Day / \$1M	Best Day / \$1M	Positive Days
final	BTCUSDT	Probability budget	7	7.69	5.58	8.5	0.22	27.54	100%
final	BTCUSDT	Direct economic	7	4.71	1.47	8.66	0	25.7	71.4%
final	BTCUSDT	Hurdle economic	7	9.25	7.38	10.62	-1.88	33.57	85.7%
validation	BTCUSDT	Probability budget	7	15.21	10.22	13.54	2.35	47.15	100%
validation	BTCUSDT	Direct economic	7	11.47	1.57	16.2	-0	48.42	85.7%
validation	BTCUSDT	Hurdle economic	7	16.8	11.64	15.29	2.63	52.26	100%
final	ETHUSDT	Probability budget	7	22.14	20.31	16.42	2.35	58.73	100%
final	ETHUSDT	Direct economic	7	16.58	13.41	18.39	0.22	58.91	100%
final	ETHUSDT	Hurdle economic	7	20.08	15.96	14.77	5.1	54.22	100%
validation	ETHUSDT	Probability budget	7	27.47	22.38	16.95	3.06	59.01	100%
validation	ETHUSDT	Direct economic	7	25.23	19.12	19.08	4.51	67.12	100%
validation	ETHUSDT	Hurdle economic	7	30.33	28.42	19.46	5.34	73.04	100%
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Paired Day Bootstrap Robustness							
Market	Comparison	Mean Daily Delta / \$1M	95% CI	Positive Days	Inference	Days	Bootstrap Samples
BTCUSDT	Hurdle VS Direct	4.54	[1.79; 7.06]	85.7%	Positive	7	5,000
BTCUSDT	Hurdle VS No Action	9.25	[2.85; 18.05]	85.7%	Positive	7	5,000
BTCUSDT	Hurdle VS Probability	1.56	[-0.3; 3.44]	85.7%	Not significant	7	5,000
ETHUSDT	Hurdle VS Direct	3.5	[-0.13; 7.47]	71.4%	Not significant	7	5,000
ETHUSDT	Hurdle VS No Action	20.08	[11.52; 32.31]	100%	Positive	7	5,000
ETHUSDT	Hurdle VS Probability	-2.06	[-4.21; 0.24]	28.6%	Not significant	7	5,000
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Prediction Diagnostic: Validation VS Final									
Split	Market	Horizon, sec	Days	Mean P(markout > 0)	Mean P(markout > BE)	Expected Markout P95, bps	Maximum Expected Markout, bps	Hurdle Predicted Positive	Direct Predicted Positive
final	BTCUSDT	600	7	0.51	0.39	7.38	31.06	50%	1.1%
validation	BTCUSDT	600	7	0.51	0.42	11.53	48.21	84.9%	7.8%
validation	BTCUSDT	120	7	0.52	0.32	5.88	23.66	25.2%	1.4%
validation	BTCUSDT	300	7	0.51	0.39	8.96	58.62	53.5%	4.6%
final	ETHUSDT	600	7	0.51	0.43	11.03	42.37	95.4%	6.4%
validation	ETHUSDT	120	7	0.52	0.4	9	28.07	63.8%	4.8%
validation	ETHUSDT	300	7	0.51	0.44	11.43	36.56	92.9%	9%
validation	ETHUSDT	600	7	0.51	0.45	17.98	58.01	99.5%	30.9%
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Oracle Feasibility within the Deployable Horizon Range							
Market	Horizon, sec	Budget	Oracle Mean / \$1M	Robust Score / \$1M	95% CI	Positive Days	Events Above BE
BTCUSDT	120	10%	26.29	20.99	[18.85; 34.43]	100%	30.8%
BTCUSDT	300	10%	38.53	32.52	[29.42; 47.36]	100%	37.6%
BTCUSDT	600	10%	54.19	44.87	[40.86; 68.13]	100%	41.5%
ETHUSDT	120	10%	46.67	35.69	[31.4; 64.41]	100%	38.3%
ETHUSDT	300	10%	67.45	53.37	[47.76; 88.57]	100%	42.7%
ETHUSDT	600	10%	94.92	74.02	[67.18; 128.31]	100%	44.9%
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Leakage And Research Governance Checklist				
Check Order	Check Name	Evidence		Status
1	Chronological split	Train, development, validation and final periods do not overlap		Passed
2	Final holdout isolation	The final period is used only after one horizon is selected on validation		Passed

3	Causal feature construction	Features use current and historical observations; future data define targets only	Passed
4	Oracle isolation	Oracle results are diagnostic and never used as deployable predictions	Passed
5	Frozen scenario assumptions	Internalization, mitigation and action cost are fixed before final reporting	Passed
6	No post-final retuning	The research_v1_0 configuration is frozen after the final holdout is viewed	Passed
7	Artifact traceability	Experiment ID, Git commit and source JSON paths are stored in ClickHouse	Passed
8	External validity	Public Binance flow is a market proxy, not proprietary bank client flow	Limitation
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Validation conclusion

The 600-seconds configuration was selected on validation before the final holdout was evaluated.

The final holdout preserved the positive direction of the validation result for both markets. Daily bootstrap intervals are positive relative to no action, while superiority over every alternative policy isn't established for every market.

The final result is therefore evidence of positive scenario-adjusted out of sample value, not evidence of guaranteed realized bank profit.